

EUGENE L. BUCHANAN

Cyber Security Specialist — AI Security Automation & Compliance



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Professional Summary

Cyber security specialist with 27 years of software and systems engineering experience, now focused on AI-driven security automation, code auditing, and compliance. Contributed security architecture review (PR #7946, #8966, #8337) to ZeroClaw (32.2k★ Rust gateway). Deployed AI agents with /bug-reaper skill for automated vulnerability hunting across 18 CWE classes with evidence-based PoC validation. Led SOC 2 compliance evidence collection at Knowledgecity. Built test automation and CI/CD security pipelines on AWS/Postgres/React. Committed to deny-by-default, principal isolation, and adversarial FP elimination.

Security Expertise



Security Domains & Tooling

SOC 2 ISO 27001 Compliance Evidence Trust Service Criteria Change Management Access Controls

Incident Response

Bug-Reaper AI Vulnerability Hunting PoC Validation CVSS Scoring WAF Bypass Bug Chaining

Recon Methodology

SAML OAuth OIDC SSH Challenge-Response Peercred Deny-by-Default Principal Isolation

SQL Injection XSS CSRF SSRF IDOR RCE Auth Bypass Data Exposure API Security

Rust Python PostgreSQL AWS (ECS, S3) Elasticsearch JSON-RPC MCP Protocol REST APIs

Docker Git

Experience

Cyber Security Engineer / AI Security Automation Contributor

2026 – Present

ZeroClaw Labs (github.com/zeroclaw-labs/zeroclaw) · Rust · 32.2k★

- PR #7946 (MERGED) — feat(runtime): context window visibility across TUI/gateway/CLI (+1,109/-22, 25 files). Established context_window as single source of truth in provider config for 9 LLM providers, preventing silent token overflow that could truncate security audit output. Added doctor command for config integrity verification.
- PR #8966 (OPEN) — feat(agent): live provider identity on usage events (+2,742/-141, 20 files). Fixed a frozen 32k context-meter stub on 1M-token models that could mask incomplete security analysis. Separated trim budget from model capacity. Added per-provider usage breakdown for cost attribution — critical when routing security scans across multiple LLMs.
- PR #8337 (OPEN) — feat(observability): herdr agent reporting (+1,712/-17, 13 files). Observer reports agent lifecycle (idle/working/blocked/released) via JSON-RPC over UDS. Approval-gate events drive blocked state for supervised tool calls — security audit trail for agent actions. Zero-config env detection, crash recovery via pane.release_agent.
- Built herdr-mcp MCP server (github.com/eugeneb50/herdr-mcp) providing pane control and inter-agent communication for multi-agent security orchestration over MCP protocol.
- Deployed /bug-reaper AI agent skill (github.com/shaniidev/bug-reaper) for automated vulnerability hunting: 18 CWE classes, 4-phase recon, exploit validation with working PoC, false-positive elimination, CVSS scoring, WAF bypass (15 products), bug chaining (P3→P1). Targets HackerOne, Bugcrowd, Intigriti, YesWeHack triage standards.

Senior QA Engineer / Test Automation Lead / SOC 2 Compliance Lead

Dec 2020 – Aug 2025

Knowledgecity LLC · Remote · AWS / Postgres / React

- SOC 2 Type II compliance: collected and organized evidence for security audits across access controls (SAML/SSO), data encryption, network security, change management, and incident response. Mapped CI/CD pipeline controls to SOC 2 trust service criteria (CC1–CC7). Worked with auditors to verify control effectiveness.
- Security testing across integration surfaces: SAML/OAuth/SFTP authentication, REST API payload validation, SQL injection prevention, XSS/CSRF verification, and data exposure audits for integrations with SAP, Oracle, Workday, UKG, Coursera, Zoom.
- Built CI/CD security pipeline with Cypress, Selenium, Postman, JUnit, Elastic, S3 — automated regression suites, health dashboards, alert systems covering security, performance, and API integrity across multi-environment deployments.
- Applied AI prompt engineering and chatbot security: validated LLM-powered tooling for prompt injection resistance, data leakage prevention, and access boundary enforcement.
- Conducted code reviews and security audits across integration, backend, API, database, and frontend surfaces. Reduced technical debt and security vulnerabilities.
- Mentored and trained test automation team in security testing best practices.

Senior Software Test Engineer IV

1997 – 1999

Microsoft · Redmond, WA · Windows Media Server

- White-box testing and security analysis for Windows Media Server, Windows 98/NT — building evaluation frameworks for complex system-level security testing.
- Deep debugging across server infrastructure, network protocols, and media codecs — identifying attack surfaces and verifying access controls.

Senior Software Engineer II / QA & Security Research

1999 – 2001

RealNetworks · Seattle, WA · Streaming Media Platform

- QA on advanced research team evaluating consumer appliances, mobile platforms, stream servers, and cellular networks — vulnerability assessment across distributed systems.
- Global presales engineering: security architecture review for enterprise deployments.

Education

- Associate Degree, Victor Valley College
- California Notary Commission (Background-Checked, Bonded)
- Toastmasters International

Key Strengths

- **AI-driven security automation.** Deployed /bug-reaper agent skill for 18 CWE classes with PoC validation, targeting HackerOne/Bugcrowd triage standards.

- **Compliance evidence.** Led SOC 2 Type II evidence collection at Knowledgecity — mapped CI/CD controls to CC1-CC7 trust criteria.
- **Security architecture review.** Reviewed 2,742-line provider identity PR and 1,712-line observability PR for ZeroClaw — deny-by-default, isolation, audit trails.
- **Auth security.** Tested SAML/OAuth/SFTP auth across 6 enterprise integrations; reviewed SSH challenge-response and OIDC for ZeroClaw.
- **27 years systems engineering.** Microsoft, IBM, RealNetworks — deep debugging across protocols, infrastructure, and access controls.
- **Mentoring & training.** Trained test automation team in security testing best practices; conducted code reviews and security audits.